

DYLAN DSOUZA

San Diego, CA

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EDUCATION

University of California San Diego

Bachelor of Science in Data Science – GPA: 3.92 (Provost Honors)

San Diego, CA

Expected March 2027

EXPERIENCE

Data Engineer Intern

Jul 2026 – Aug 2026

Capgemini

Mumbai, India

- Built a serverless AWS RAG pipeline for PDF Q&A using S3 storage, Titan embeddings via Bedrock, FAISS indexing, Lambda-parallelized retrieval, and Claude Sonnet responses via API Gateway — deployed to EC2 using CodePipeline.
- Completed a 15-course generative AI certification program covering prompt engineering, MCP protocol, agentic AI workflow design, RAG and vector database architectures, and AI governance.

AI Engineer

Oct 2025 – Jun 2026

Computer Science & Engineering Society

San Diego, CA

- Implemented a multi-stage Neo4j retrieval pipeline fusing MiniLM vector search and BM25 via reciprocal rank fusion, achieving 89% recall and 84% precision for an agentic AI platform modeling professional decision-making styles.
- Constructed a knowledge graph pipeline extracting LLM-generated concepts from research documents, mapping entity relationships via Girvan-Newman community detection across 8+ entity categories, and built the React/Vite frontend.

Machine Learning Engineer

Oct 2025 – May 2026

Engineers for Exploration

San Diego, CA

- Engineered a Google Earth Engine pipeline staging Sentinel-2 optical bands fused with 64-band AlphaEarth embeddings across training tiles in Madagascar and Mozambique, restricting downloads to mangrove regions via ESA WorldCover.
- Fine-tuned a continual learning segmentation model on 21,000+ GeoTIFF image-mask pairs, achieving 82–85% mIoU across 4 land-cover classes for global-scale pixel-level mangrove mapping.

Data Analyst

May 2025 – May 2026

Scripps Institution of Oceanography

San Diego, CA

- Automated 11 years of coral reef data processing (60% cut in manual processing time), building a Python ETL pipeline that standardized 40,000+ benthic annotations into clean datasets for downstream analysis across 15 reef sites.
- Applied EDA and time series analyses across 5 experimental groups and 16 survey time points, finding active restoration plots reached 30–40% coral cover versus 13% in controls — quantifying the long-term impact of transplantation.

PROJECTS

Who Gets the Old Planes? | Personal Project

[View Project](#) | [GitHub](#)

- Cross-referenced 2.7M flights with FAA aircraft registration records to map narrowbody fleet age across U.S. airports, uncovering an 18-year disparity between the oldest and youngest fleets driven by market competition, not airline.

DegreeDelta: Quantifying Education Returns | Course Project (Statistical Methods)

[View Project](#) | [GitHub](#)

- Quantified the college income premium across all 50 U.S. states using 3.4M weighted ACS PUMS records, revealing that cost-of-living adjustment dramatically reshuffles which states actually deliver the highest return on a degree.

ENSOCast: Climate Analytics Dashboard | Conference Project (SDUTC 2025)

[View Project](#) | [GitHub](#)

- Engineered a Streamlit climate dashboard on 40+ years of NOAA data; benchmarked Random Forest, XGBoost, 1D CNN, LSTM, and Ensemble models achieving 82% accuracy in 3-class ENSO phase prediction.

Predicting U.S. Power Outage Severity | Personal Project

[View Project](#) | [GitHub](#)

- Classified U.S. power outage duration as long or short across 16 years of continental data (2000–2016), engineering geographic, climatic, and infrastructure features to achieve 80% accuracy with a tuned Random Forest.

SKILLS

Languages: SQL (PostgreSQL), Python, R, JavaScript, HTML, CSS, Java

Libraries: pandas, NumPy, statsmodels, SciPy, Matplotlib, Seaborn, scikit-learn, XGBoost, NLTK, HF Transformers, FAISS

Frameworks: Streamlit, dbt, pytest, MLflow, Airflow, Spark, PyTorch, LangChain, FastAPI

Tools: Tableau, Power BI, Excel, BigQuery, Snowflake, Databricks, Jupyter, Git, AWS, GCP, Docker, MongoDB, Neo4j, Bash

Core Competencies: EDA, Statistical Modeling, Hypothesis Testing, Data Visualization, Feature Engineering, RAG, GenAI